

Online Retail EDA – Customer & Returns Analytics

Data-Driven Insights from 500,000 Transactions (2009–2011)

Artur Melnyk – Data Analyst
Python · Excel · Tableau

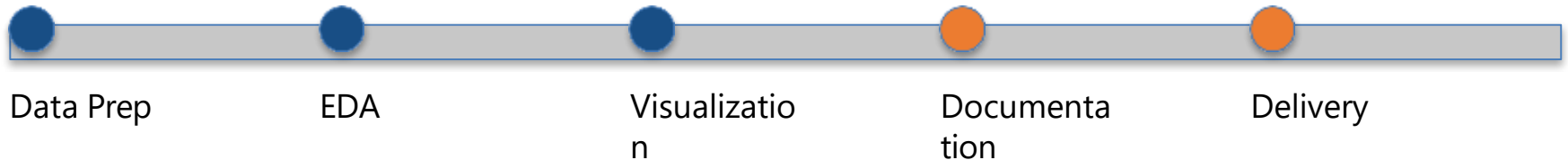
Project Overview

- Objective: Uncover revenue, customer, and return trends from Online Retail II dataset.
- Scope: Period 2009–2011 · ~500K transactions · 8 columns

Business Questions:

- What drives revenue growth?
- Who are top customers and high returners?
- Which products/countries contribute most?

Timeline: Data Prep → EDA → Visualization → Documentation → Delivery



Tools & Methodology

- Data Preparation – Python (Pandas, NumPy) – Handle nulls, clean types
- Exploration – Matplotlib, Seaborn – Identify sales & return trends
- Export for Viz – Excel – Prepare summary tables
- Visualization – Tableau – Build interactive dashboards
- Documentation – PDF & README – Ensure reproducibility

Data Cleaning Challenges & Solutions

- Mixed Date Formats – `pd.to_datetime(errors='coerce')`
- Missing Customer IDs – Dropped / flagged
- Currency Strings – Converted to float
- Invalid Descriptions – Flagged as Return Reasons
- Lesson: Data cleaning = foundation of credible analysis

KPI Summary Bar

-  Total Revenue: €14.7M (83% from UK)
-  Top Customer (ID 18102): €608K spent
-  Average Return Rate: 7.3%
-  Most Returned Product: Paper Craft Little Birdie
-  Countries Covered: 38 (UK & EIRE dominate)

Sales & Revenue Performance

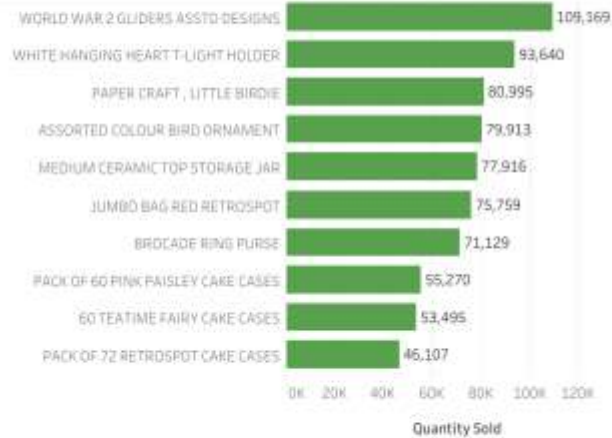
Online Retail EDA Story

Sales & Revenue
Performance

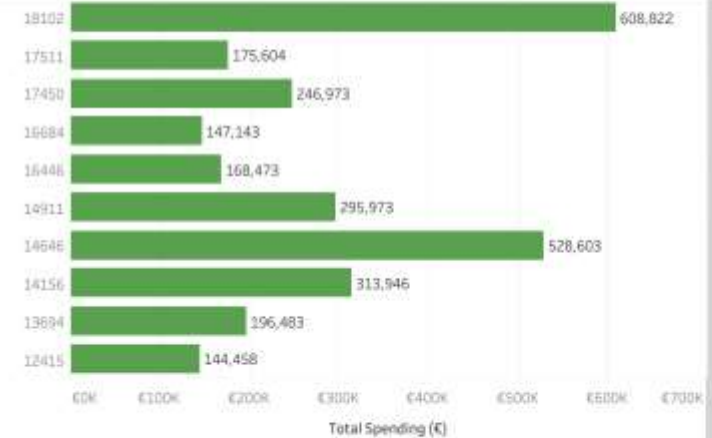
Customer & Returns Analysis

Online Retail Sales Performance Overview

Top Products by Quantity Sold



Top 10 Customers By Revenue

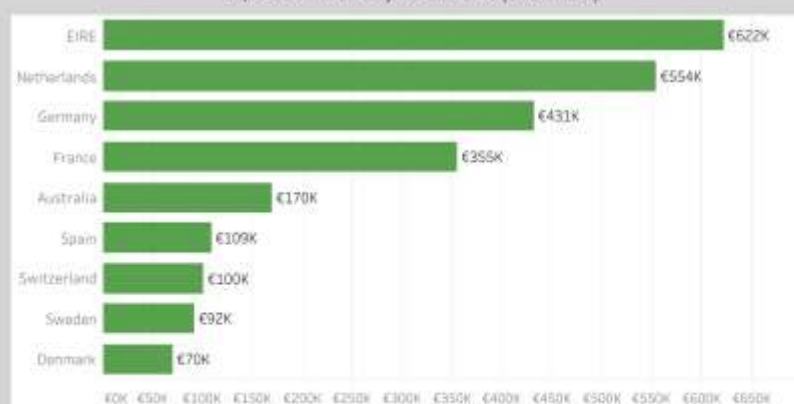


- Country
- ☐ Australia
 - ☐ Denmark
 - ☐ EIRE
 - ☐ France
 - ☐ Germany
 - ☐ Netherlands
 - ☐ Spain
 - ☐ Sweden
 - ☐ Switzerland
 - ☒ United Kingdom

Monthly Revenue Trend



Top Countries by Revenue (Excl. UK)



Customers & Returns Analysis

Online Retail EDA Story

Sales & Revenue
Performance

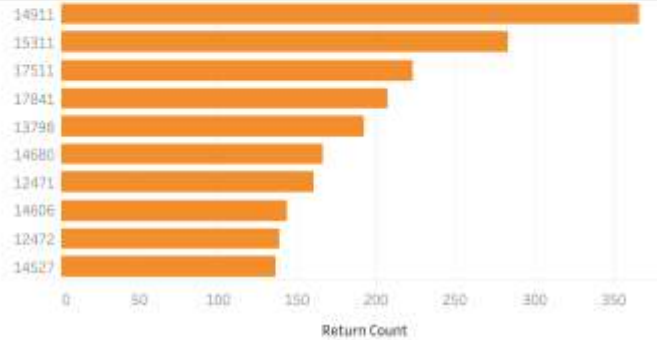
Customer & Returns Analysis

Customer & Returns Analysis

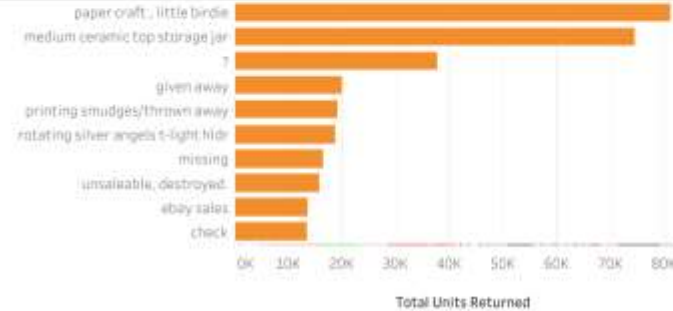
Month (MR)

- ☒ January
- ☒ February
- ☒ March
- ☒ April
- ☒ May
- ☒ June
- ☒ July
- ☒ August
- ☒ September
- ☒ October
- ☒ November
- ☒ December

Top Returning Customers



Most Returned Products/Return Reasons



Note: Some values represent return reasons, not product names.

TOP RETURNING CUSTOMER

Customer ID: 14911
Returns: 366 Units

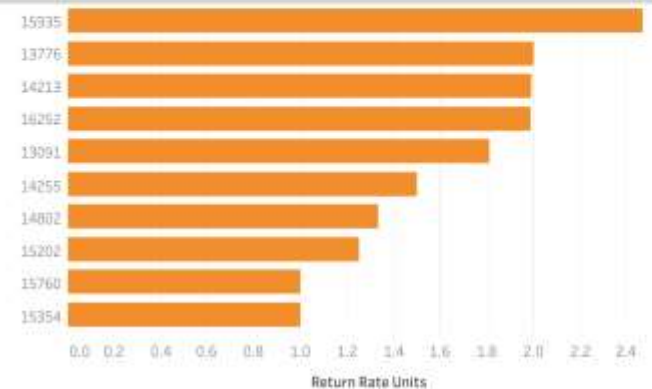
TOTAL UNITS RETURNED

310,150

Monthly Return Trend



Customers with Highest Return Rates



Top 3 Insights

1. GB" UK accounts for >80% sales → Focus marketing/logistics on UK
2. 5% customers → 60% revenue → Loyalty & retention programs
3. Volume returners $\neq$ Rate returners → Differentiate return policies

Lessons Learned

1. Validate categorical semantics → detect hidden return reasons
2. Verify data types before Tableau import
3. Curate Top N → avoid clutter
4. Document continuously → reproducibility
5. Separate return metrics → volume vs rate clarity

Deliverables & Next Steps

- Deliverables: Notebook, Clean Excel, Tableau Dashboard, PDF Documentation, README
- Dashboard Link: Tableau Public (EDA Visualization)
- Next Steps: Extend to RFM segmentation, automate monthly return alerts

Summary & Closure

- Complete EDA case: raw data → insights → dashboards → documentation.
- Business-ready: actions tied to insights (UK focus, loyalty, differentiated returns).
- Reproducible workflow: notebook + cleaned dataset + Tableau + PDF + README.
- Executive communication: KPI slide, curated Top-N visuals, branded footer.
- Next: RFM segmentation, CLV model, automation of monthly return alerts.
- *From Data to Decisions — Artur Melnyk, 2026*

GitHUB: <https://github.com/ArturMelnik-analyst/EDA-onlineretail>

Tableau:

<https://public.tableau.com/app/profile/artur.melnik/viz/EDAVisualization/OnlineRetailEDAStory>